

Clinical Case Notes — 19-Year-Old with Acute Pleuritic Chest Pain

Teaching case documentation · Urgent Care / Emergency Department · 2026

CHIEF COMPLAINT

Sharp left-sided chest pain for two days.

History of Present Illness

The patient is a 19-year-old male college student who presents with two days of sharp, stabbing, left-sided chest pain. He describes the pain as "like someone's stabbing me on the left side." The pain is distinctly **pleuritic** — markedly worse with deep breaths — and **positional**: worst when lying flat on his back, substantially improved when sitting up and leaning forward. He reports mild radiation to the left trapezius ridge and neck. He endorses a low-grade fever and subjective chills starting yesterday, plus a dry, non-productive cough.

One week prior to presentation he had a viral upper respiratory illness: sore throat, rhinorrhea, and diffuse myalgias that resolved over three days without treatment. He denies any chest trauma or heavy lifting. He denies exertional chest pain, exertional syncope, palpitations, or shortness of breath at rest. No history of similar pain. He is a varsity soccer athlete and is concerned about clearance to play next week.

PAST MEDICAL HISTORY · MEDICATIONS · SOCIAL

- No past medical history. No prior surgeries.
- No medications, no known drug allergies.
- Non-smoker, no alcohol or drug use. Varsity college soccer player (midfielder), trains 5 days per week.
- No family history of sudden cardiac death, cardiomyopathy, or premature coronary disease.

VITAL SIGNS

Temp	HR	BP	RR	SpO2
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37.9°C (100.2°F)	102 bpm	118/74 mmHg	18	97% on room air
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Physical Examination

General: Young adult male, uncomfortable, repeatedly shifting position on the gurney; prefers sitting upright and leaning forward on the exam table.

Neck: JVP not elevated. No lymphadenopathy. No Kussmaul sign.

Cardiac: Regular tachycardic rhythm. A scratchy, high-pitched friction rub is audible at the left lower sternal border with the patient sitting forward at end-expiration; the rub has two distinct components and varies with position. Heart sounds otherwise normal; not distant or muffled. No pulsus paradoxus appreciated (≤ 10 mmHg by cuff). No gallop.

Pulmonary: Clear to auscultation bilaterally; no rales, no wheeze. Pain reproducible with inspiration but no reproducible tenderness on chest wall palpation (argues against musculoskeletal etiology).

Extremities: Warm, well perfused. No edema, no unilateral leg swelling or tenderness.

Electrocardiogram

Sinus tachycardia at 102 bpm. **Diffuse concave (upsloping) ST-segment elevation** in leads I, II, aVL, and V2–V6. **PR-segment depression** in the same leads. Reciprocal changes confined to lead aVR: ST depression with PR elevation (the "aVR sign"). No reciprocal ST depression in other territories. No Q waves. No PR prolongation. Intervals otherwise normal. Impression read by attending: "Diffuse ST elevation with PR depression and aVR reciprocal changes — pattern most consistent with acute pericarditis (Stage I). No territorial ST-elevation pattern; cath lab activation not indicated."

LABORATORY RESULTS

Test	Result	Reference
WBC	$11.2 \times 10^9/L$	4.0–11.0
CRP	48 mg/L (elevated)	< 5

ESR	42 mm/hr (elevated)	< 20
High-sensitivity troponin I	Undetectable (×2, 4 hours apart)	< 99th percentile
BMP (Na/K/Cr/BUN)	Within normal limits	—
Hemoglobin	15.1 g/dL	13.5–17.5

Echocardiogram (Bedside, Followed by Formal Study)

Normal biventricular systolic function, LVEF 58%. No regional wall motion abnormalities. **Small circumferential pericardial effusion** (posterior 8 mm), without echocardiographic signs of tamponade: no right atrial collapse, no right ventricular diastolic collapse, no IVC plethora, no respiratory variation of inflow velocities. No valvular pathology. Aortic root normal.

Assessment

Acute idiopathic (post-viral) pericarditis with small pericardial effusion.

The patient meets all four diagnostic criteria: typical pleuritic, positional chest pain (better leaning forward, worse supine); pericardial friction rub on auscultation; stage I ECG changes (diffuse concave ST elevation with PR depression and the aVR sign); and a new small pericardial effusion on echocardiogram. Supporting features: elevated CRP and ESR, low-grade fever, and a viral upper respiratory illness one week prior.

Differential considered and excluded: acute coronary syndrome (age, risk profile, non-territorial ECG, negative serial troponins); myopericarditis (negative troponins, normal systolic function); pulmonary embolism (no risk factors, clear lungs, normal SpO₂, no tachycardia disproportionate to fever); pneumothorax (normal lung exam, no trauma); musculoskeletal pain (no palpation tenderness, friction rub present).

Plan

- **Ibuprofen 600 mg PO three times daily** × 7–10 days, then taper over 2 weeks as symptoms resolve
- **Colchicine 0.5 mg PO twice daily** (patient weighs 78 kg) × **3 months** — reduces recurrence risk

- Pantoprazole 40 mg daily for gastric protection while on NSAID therapy
- **Activity restriction:** no competitive sports or strenuous training for a minimum of 3 months; requires symptom resolution, CRP normalization, stable ECG, and resolution of the effusion on follow-up echo before graded return — per sports cardiology guidance
- Return precautions reviewed: worsening dyspnea, syncope, palpitations, fever persisting beyond 5–7 days, or escalating pain
- Cardiology follow-up in 1 week with repeat CRP; repeat echocardiogram in 2–4 weeks to document effusion resolution
- Anticoagulants avoided. No corticosteroids — not first-line, and associated with higher recurrence

TEACHING NOTE FOR FELLOWS

Every element of this case maps to the diagnostic criteria and treatment principles in the accompanying clinical reference. The trap this case is built around: a young patient with "ST elevation on ECG" can be mistakenly routed to cath-lab activation. Note the ECG pattern (diffuse + concave + PR depression + aVR changes), the positional pain history, the friction rub, the negative troponins, and the recent URI — the full pattern makes the diagnosis. Also note what the plan deliberately avoids: steroids, anticoagulation, and premature sports clearance.

Teaching case documentation prepared for APP fellowship education. Composites of typical findings; not a real patient record.